

Deepak Mallampati

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Kent, OH (US-based, F-1 OPT, open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping clinical RAG + multi-agent workflows + evaluation frameworks in HIPAA-conscious healthcare environments. Production gains include ~35% retrieval precision, >30% hallucination reduction via schema-validated multi-agent contracts, 15-20% recall lift on high-risk patient categories. Master's degree (Kent State). Strong end-to-end ownership across data, prompts, models, evals, APIs, and Docker-based deployment.

CORE COMPETENCIES

Applied AI / LLM Engineering • Clinical AI Agents + Multi-step Orchestration • RAG Pipelines + Vector Search + Embeddings • Evaluation Frameworks + Safety + Audit Trails • HIPAA-conscious Healthcare AI • Predictive ML (scikit-learn, XGBoost) • LLM APIs (OpenAI, Anthropic, OSS) • FastAPI + Docker + Structured Logging

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee

USA | Healthcare Technology, AI Consulting, SaaS Healthcare

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking + transformer embeddings improved retrieval precision ~**35%** and reduced irrelevant retrieval >**30%**; retrieval-grounded response alignment exceeded **90%** in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; ~**25%** stability improvement.
- Built **evaluation pipelines + safety + audit trails**: retrieval-grounded alignment evaluator, de-identification gates, data-lineage docs, stakeholder-facing system-limitation docs - clean clinical sign-off.
- Shipped predictive ML pipelines (scikit-learn, XGBoost) for patient no-show + care engagement scoring; **15-20% recall** gains on high-risk categories holding precision >90%.
- Packaged ML/LLM inference as **FastAPI/Flask + Docker** with structured logging + load simulation; reduced post-deployment defects ~**30%**.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India | Oil & Gas, Enterprise ERP

- Optimized T-SQL stored procedures (Synthesis Order-to-Cash); reduced query execution time ~**20%** and latency ~**25%**.
- Built Jenkins CI/CD pipelines for schema + stored procedures; reduced deployment errors >**30%**, release cycle ~**35-40%** faster.
- Supported RBAC + audit logging for financial modules in compliance-sensitive oil & gas - discipline transfers to HIPAA contexts.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India | Nonprofit Tech, Video Analytics, Applied NLP

- Document Q&A with RAG (semantic chunking + embedding retrieval); ~**30%** hallucination reduction; >85% contextual answer accuracy in controlled tests.
- Transformer-based video summarization across 5,000+ sessions; **60-70%** review-time reduction; ~85% highlight alignment with human curation.
- Deployed as Flask API with web UI for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. **RAG-grounded CPT/ICD validation** against vector-indexed policy documents; ANN similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped, Chrome Web Store

Full-stack Chrome MV3 extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-provider abstraction layer + payload handling + caching + privacy policy. Demonstrates production multi-LLM provider orchestration.

Agentic Pixel Character Synthesis & Animation Engine ongoing

LLM-based agent orchestrator decomposes high-level prompts into generation tasks; Stable Diffusion + LoRA for identity consistency, ControlNet + OpenPose for pose conditioning. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript app coordinating multimodal APIs. **Intermediate structured prompt transformation layers** improved contextual alignment ~30% and first-pass image success rate ~25-30%.

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN with unified YOLOv8 detect+classify; ~30% latency reduction; sliding-window confidence aggregation cut blink-driven false positives ~25%.

EDUCATION

Master's Degree - Kent State University, Ohio, USA Aug 2023 - May 2025

WORK AUTHORIZATION

F-1 OPT - authorized to work in the United States now; future sponsorship required.

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP

LLM, GenAI & Agentic: RAG, agentic LLM workflows, multi-step agent orchestration, schema-validated JSON contracts, structured outputs, prompt engineering, tool calling, evaluation pipelines, safety tests, regression suites, guardrails, grounding

Vector Search & Embeddings: Transformer embeddings, recursive semantic chunking, vector indexing, retrieval optimization, similarity search for clinical knowledge bases

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Fine-tuning & Multimodal: Stable Diffusion fine-tuning + LoRA (identity consistency), ControlNet + OpenPose/MediaPipe pose conditioning, YOLOv8

Healthcare AI: HIPAA-conscious governance, de-identification, data lineage, evaluation audit trails, system-limitation docs, CPT/ICD policy retrieval

Infra & DevOps: FastAPI, Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, structured logging, observability

Domains: Clinical AI, Applied AI, Healthcare AI, Multi-agent Systems, Enterprise AI